layer flooded with electrolyte. The GDL layer is mounted on a scaffolding made of a rigid porous layer of carbon cloth, metal foam, or porous PTFE to improve structural stability, porosity, hydrophobicity, and conductivity. The catalyst layer is composed of catalyst powder pressed together with conductive carbon powder, which is cheap but has poor corrosion resistance at strongly oxidising positive potentials. Alternatively, an air electrode can be made where both GDL and the catalyst are nickel-based in the form of spinel-catalysed nickel powder and nickel foam to achieve good stability and current density as high as 100mA cm^{-2} at a recharging potential of +0.55V versus Hg/HgO.

The overall thickness of the air electrode has a large influence on cell performance. A gas diffusion electrode should have a thickness of 0.1mm maximum, typically 100–400µm, to reduce the time needed for oxygen to diffuse through the GDL. A very thin catalyst layer allows the OH- in the electrolyte to diffuse more rapidly towards the catalyst during oxygen reduction and allows the oxygen bubbles to escape from the catalyst during oxygen evolution. The corrosion of carbon-based GDLs may cause the degradation of the electrodes, reducing cycle life. Hence, metal-based materials like Ni foam and stainless steel/titanium mesh are preferred as media for gas diffusion.

Composite air electrode preparation

The composite air electrode consists of an electrocatalyst layer, a current collector of Ni mesh, and a PTFE layer, which are sandwiched together. The catalyst layer is prepared by mixing all ingredients like perovskites, graphite, and spinel in a solvent in the required proportion and then blending in a blender rotating at 6000rpm. The blended solution is filtered, and the paste of the catalyst is rolled onto a 100 mesh Ni screen. The pasted electrode is pressed at 375kg/cm^2. The other side of the electrode is then painted with PTFE solution. The assembly is then sintered at 325°C in an N2 atmosphere. The PTFE layer acts as a waterproof gas-diffusing layer (GDL), providing larger gas transport pores, which contribute to a fast and uniform supply of oxygen while preventing the leakage of electrolyte. The electrocatalyst layer can also be formed by electrodeposition of catalyst material on carbon cloth. Thereafter, the gas diffusion side is provided with a rigid porous layer of carbon cloth or metal foam by cold press.

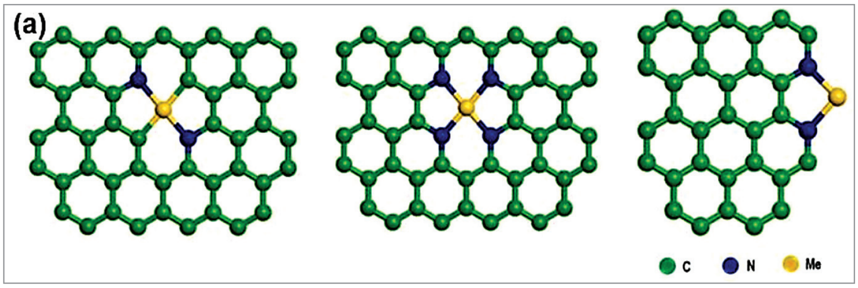

Fig. 11: Nanostructured M-N-C molecules of bifunctional catalyst

Bifunctional catalyst

Charging and discharging reactions in Fe-air batteries involve oxygen evolution reactions (OER) and oxygen reduction reactions (ORR). The ORR at the air electrode involves several steps:

1. Oxygen from the atmosphere is brought to the catalyst surface, where it is absorbed on the active site.

2. Electrons from the catalyst are then transferred to oxygen, resulting in weakening and removal of the oxygen bond.

3. Finally, the catalyst surface releases hydroxide ions to the electrolyte.

OER is the exact opposite of ORR. However, these two reactions are slow, and a large overpotential is required to overcome this sluggishness. This means there is a voltage loss that reduces the capacity utilisation of the cell.

To offset this problem, active bifunctional electrocatalysts like noble metals, perovskites, spinels, and non-noble metal catalysts are required. Noble metal catalysts like Pt, Pd, and Pt alloys for ORR, and RuO2 for OER are the best, but their prohibitive cost limits their commercial use. Non-noble metal catalysts like transition metal oxides of perovskites, spinels, pyrochlores, and layered double hydroxides are preferred due to their low cost, ease of manufacture, robustness, stability, durability in alkaline electrolytes, and high bifunctional OER and ORR activity.

Spinels are AB2O4 structure compounds, where A is rare earth divalent ions like La, Ce, Sm, Nd, etc, and B is a transition metal trivalent ion like Cr, Mn, Fe, Co, Ni, etc. One such spinel, Co3O4, with its tetrahedral sites partially substituted by Ni, Mn, Cu, or Zn, and MnO2-perovskites mixed catalysts performs well for bifunctional applications in alkaline medium.

Layered double hydroxides (LDH), another important class of catalysts, are mixed valence, ionically conducting solids with a layered structure of Ni-Fe, Ni-Co, Co-Fe, Ni-Mn, Co-Ni, etc., where the interlayers are filled with carbonate, nitrate, and chloride ions. They can be easily oxidised to highly active